Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2023/2024

COURSE NAME	:	OBJECT-ORIENTED PROGRAMMING
COURSE CODE	:	BIC 20904
PROGRAMME CODE	:	BIS / BIP / BIW / BIM
EXAMINATION DATE	:	JANUARY/FEBRUARY 2024
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

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Q1 Answer question **Q1(a) – Q1(g)** based on the program codes in **Figure Q1.1**.

```
class Data1{
      private int a = 1;
      private double c = 2.456;
      public double getData() {
            return c;
      }
}

class Data2{
      private int b, c;
      public final String COLOR = "Red";
      Data2(int f, int g){
            this.b=f;
            this.c=g;
      }
      public String getColor() {
            return color;
      }
      public int sum () {
            return b + c;
      }
}

public class Main {
      public static void main(String[] args) {
            Data1 obj1 = new Data1();
            Data2[] obj2 = new Data2[3];

      // statements (a) - (g) will be implemented here

      }
}
```

Figure Q1.1

Determine whether each of the following statement is **VALID** or **INVALID** to be implemented in the main method.

(a) `System.out.println(obj1.a);`

(2 marks)

(b) `System.out.println(obj1.getData());`

(2 marks)

(c) `System.out.println(obj2.getColor());`

(2 marks)

(d)
```
obj2[0] = new Data2(1,2);
double e = 2 + obj2[0].sum();
```

(2 marks)

(e) `obj2[1].COLOR = "Blue";`

(2 marks)

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(f)
```
double d = 10 * obj1.getData();
System.out.println(d);
```

(2 marks)

(g)
```
obj2[2] = new Data2(4,6.5);
System.out.println(obj2[2].sum());
```

(2 marks)

Q2 Answer question **Q2(a) – Q2(c)** based on the information in **Figure Q2.1.**

A class named Rectangle contains:

- Two double data fields named width and height that specify the width and height of the rectangle.
- A no-argument constructor that creates a rectangle with default value 1 for both width and height.
- A constructor that creates a rectangle with the specified width and height.
- A method named getArea() that returns the area of rectangle.
- A method named getPerimeter() that returns the perimeter of rectangle.
- A toString method that returns a string contains height, width, area of rectangle, and perimeter of rectangle.

Figure Q2.1

(a) Draw the UML diagram for the `class`.

(9 marks)

(b) Write a Java program to create and implement `class Rectangle`.

(21 marks)

(c) Write a Java program to create `class MainRectangle`. Implement method `main` in this class. Instantiate two `Rectangle` objects, one with default values and the other with width 4 and height 40. Display the width, height, area, and perimeter of each rectangle using `toString()` method.

(6 marks)

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Q3 Answer question **Q3(a) – Q3(c)** based on the program code in **Figure Q3.1** and information in **Figure Q3.2.**

```
class Circle{
      protected double radius;
      protected final double PI=3.142;

      public Circle (double radius){
            this.radius=radius;
      }

      public double getArea() {
            return PI*radius*radius;
      }

}
```

Figure Q3.1

A class named Cylinder is a derived class of class Circle.
Class Cylinder contains:

- data fields named height that specify the height of the cylinder.
- A constructor that creates a cylinder with the specified radius and height.
- A method named getVolume() that returns the volume of cylinder with the formula of volume = $\pi \times radius^2 \times height$

Figure Q3.2

(a) Draw the UML diagram for the `classes` with correct relationship.

(10 marks)

(b) Write a Java program to create and implement `class Cylinder`.

(10 marks)

(c) Write a Java program to create `class MainShape`. Implement method `main` in this `class`. instantiate objects from `class Circle` and `Cylinder` with value of radius 4 and height 5. Invoke the required methods to calculate the area of circle and volume of cylinder and display the value on the output screen.

(10 marks)

- END OF QUESTIONS -

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